Global CO_2 Emissions: Who Emits, How Much, and What's Changing

Fossil fuels & industry, 1750–2024

For policy analysts, climate negotiators, and sustainability researchers

Data: Global Carbon Budget 2025 via Our World in Data

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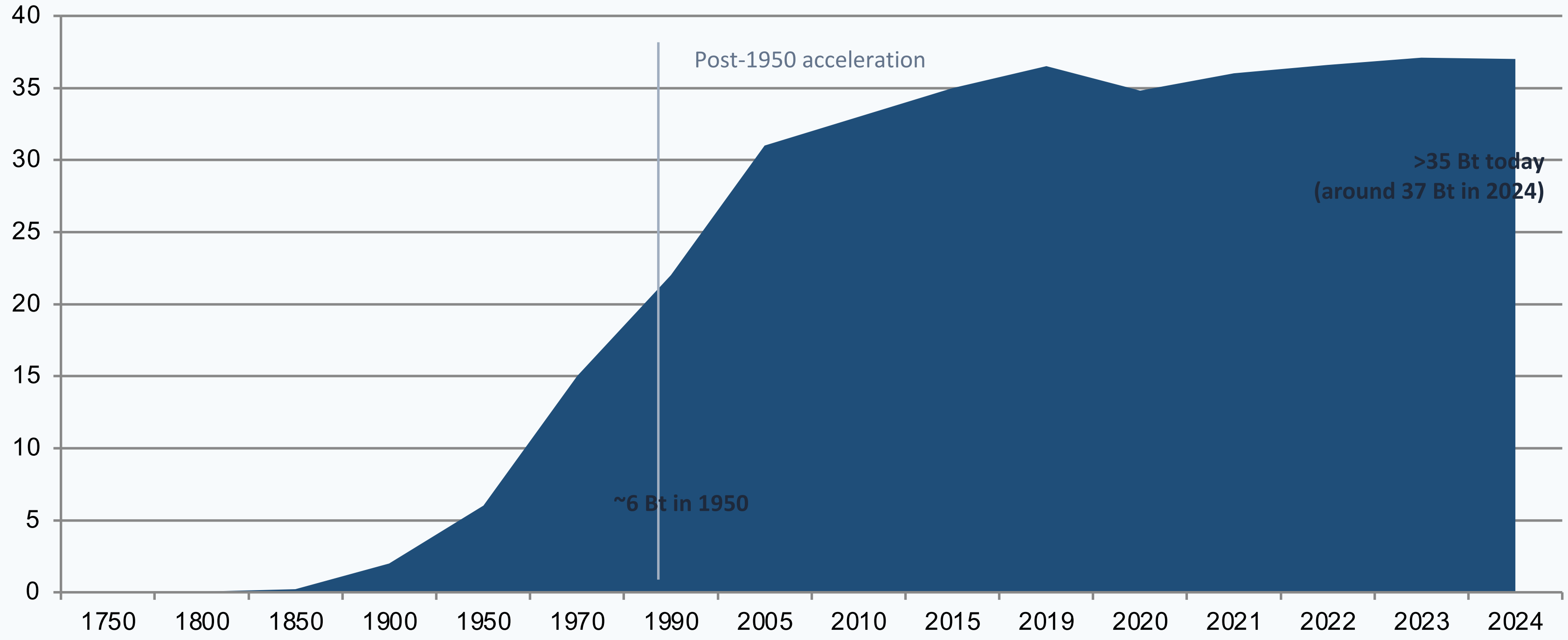
Long-run trajectory (fossil fuels and industry) shows sharp post-1950 acceleration, then slower growth without a peak.



Inflection points

- 1750–1900: very low baseline
- 1900–1950: industrial expansion
- 1950–2010: rapid scale-up
- 2010–2024: slower rise, no peak yet

Trend signal: growth has slowed in recent years, but global emissions have not yet peaked.

Today

Regional shares of annual global CO_2 emissions have shifted from Atlantic dominance to Asian growth.

Representative share snapshots (from source brief):

Year	Europe+US share	Rest of world share
1900	>90%	<10%
2024	<33%	>67%

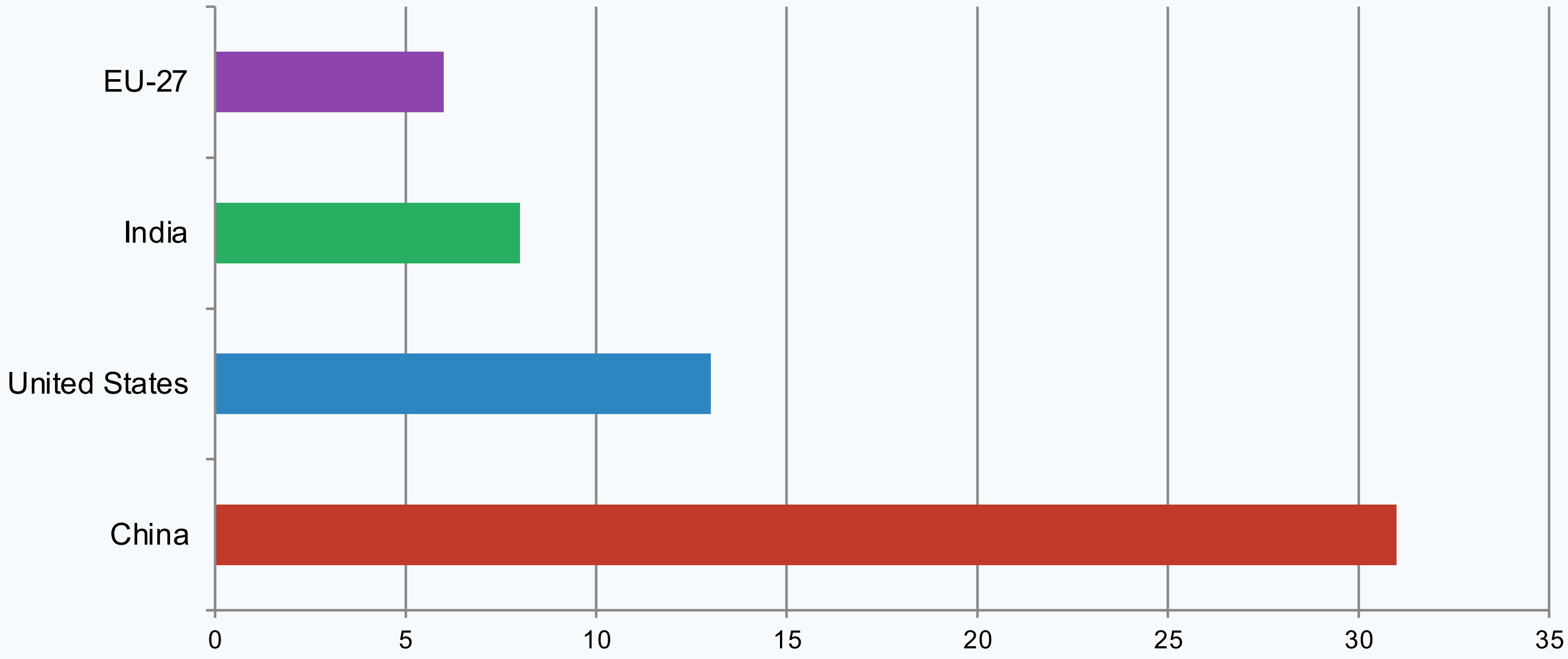
STACKED AREA CHART PLACEHOLDER
Regional share of global emissions, 1900→2024

Bands: Europe, North America, China, India, Other Asia, Other
Visual message: Europe+US bands shrink; Asia (esp. China) expands

Narrative: responsibility for current annual emissions has shifted sharply, even as historical cumulative emissions remain concentrated in earlier industrializers.

China Now Emits More Than One Quarter of Global CO_2

Country-level shares of annual emissions in 2024.



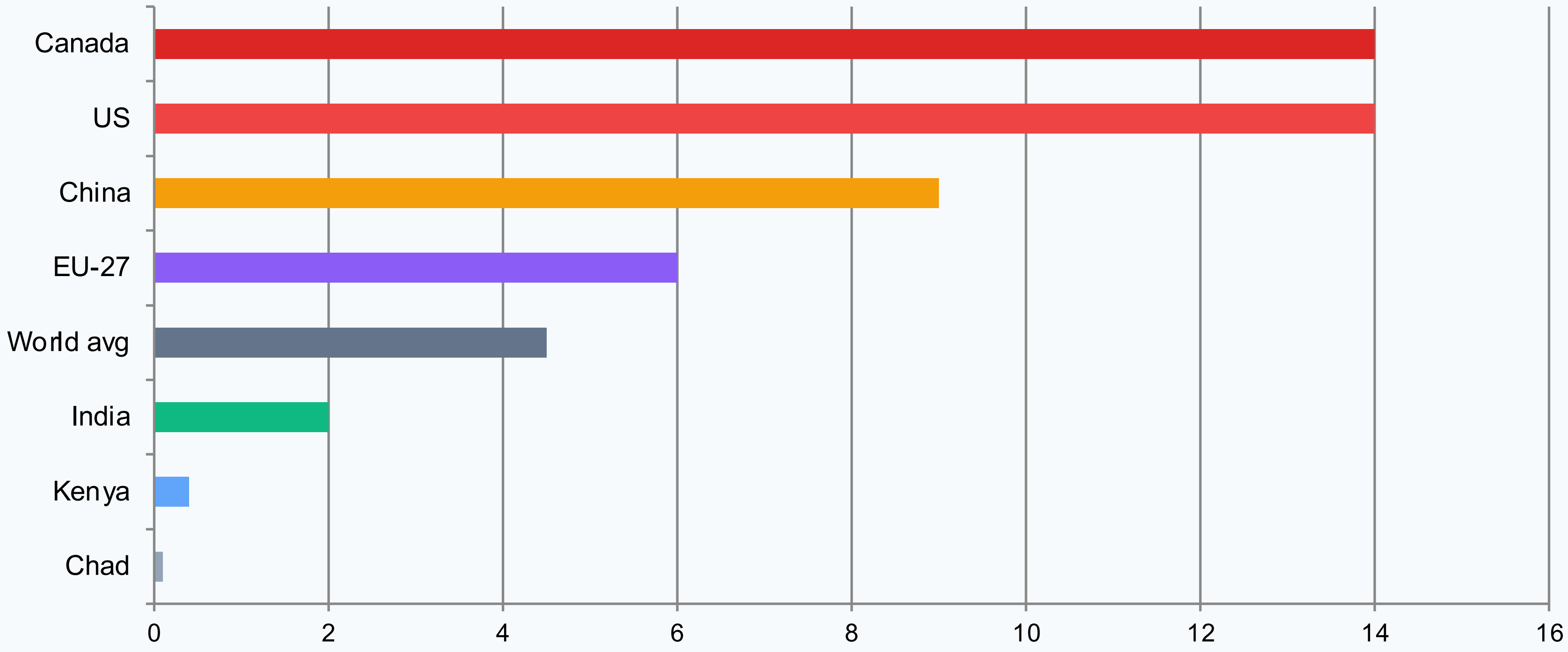
Headline insight

China: ~31%
of global annual CO_2

US: ~13%
India: ~8%
EU-27: ~6%

Canada

A ~150x gap separates the lowest- and highest-emitting nations per person.



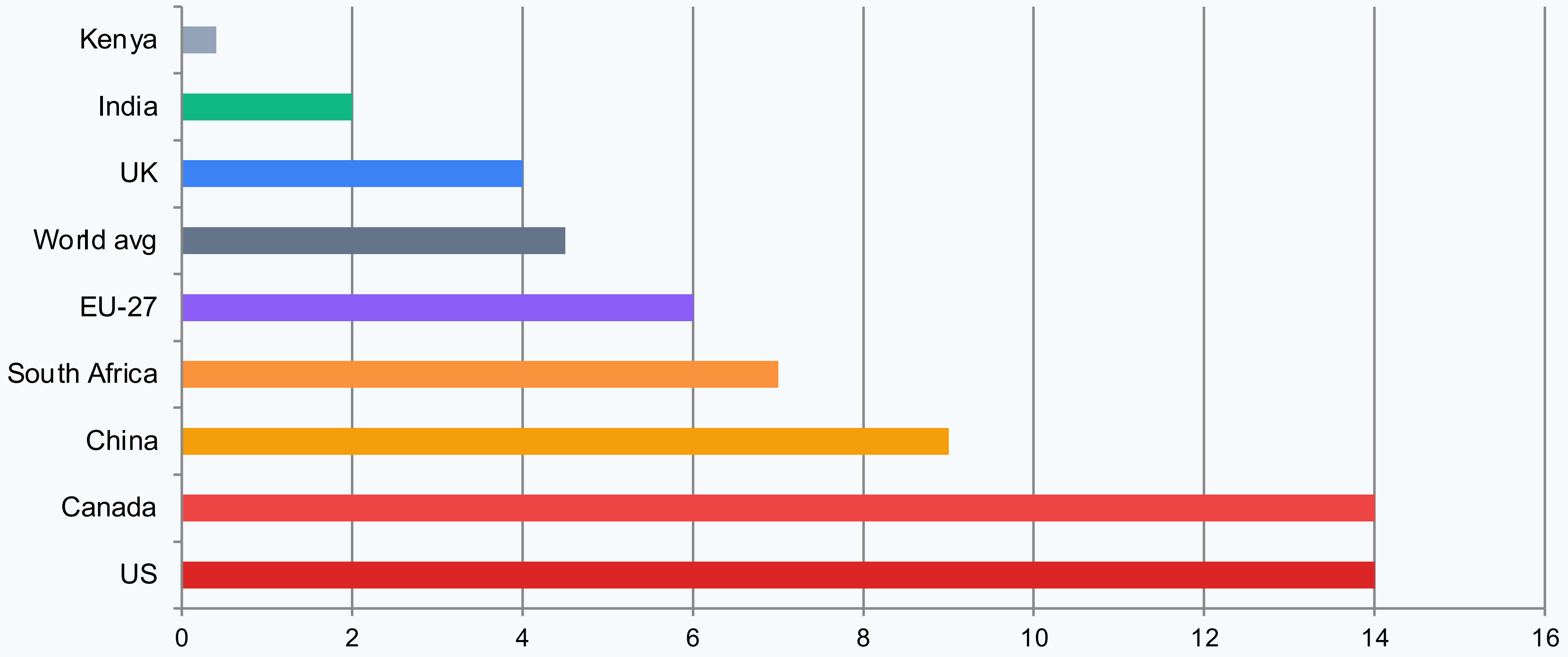
150x inequality

0.1 t (Chad) → ~14 t (US/Canada)

European peers such as France and the UK are much lower than the US despite comparable living standards.

in India

Current per-capita snapshot of key economies and benchmarks.

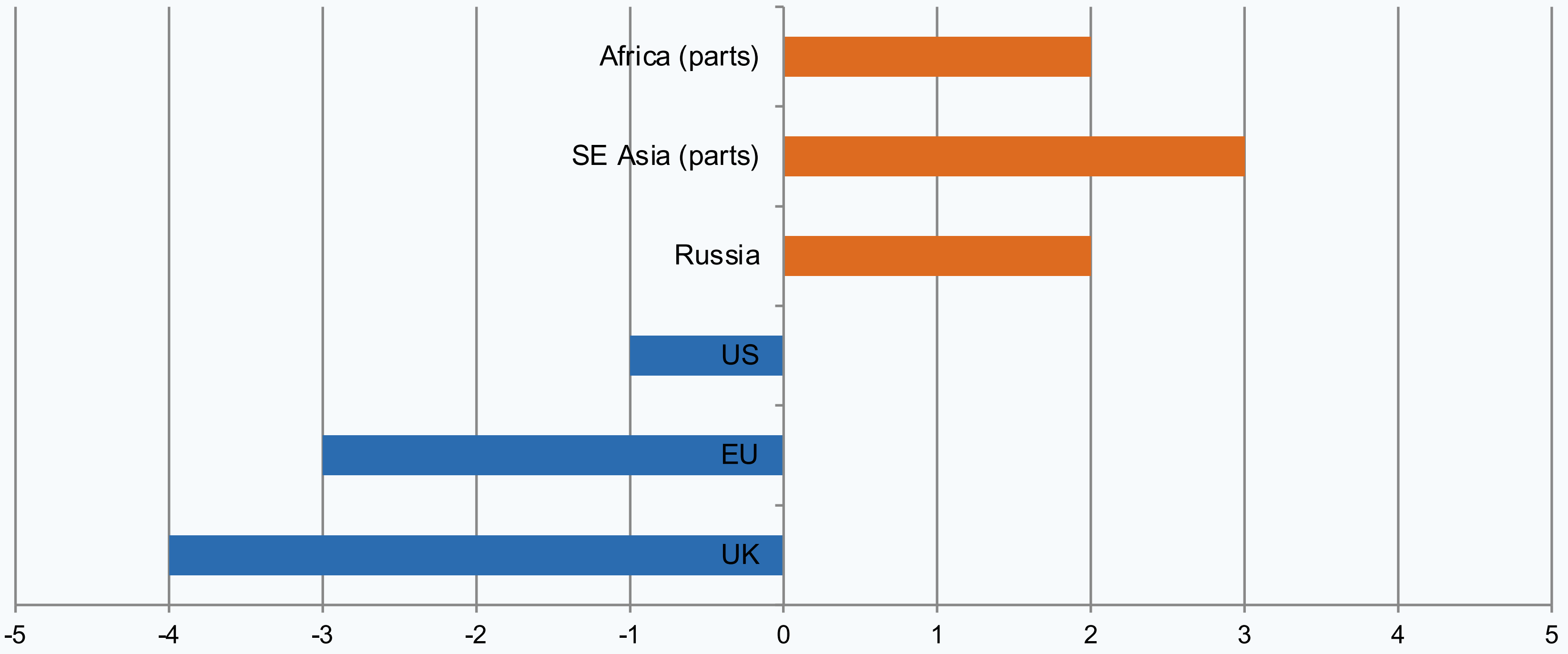


Interpretation

- China is the largest total emitter, but per person remains below the US.
- India remains far below global average.
- See source figure_8.jpg for historical evolution context.

and Africa Grew

Diverging annual changes indicate uneven progress in the latest year.



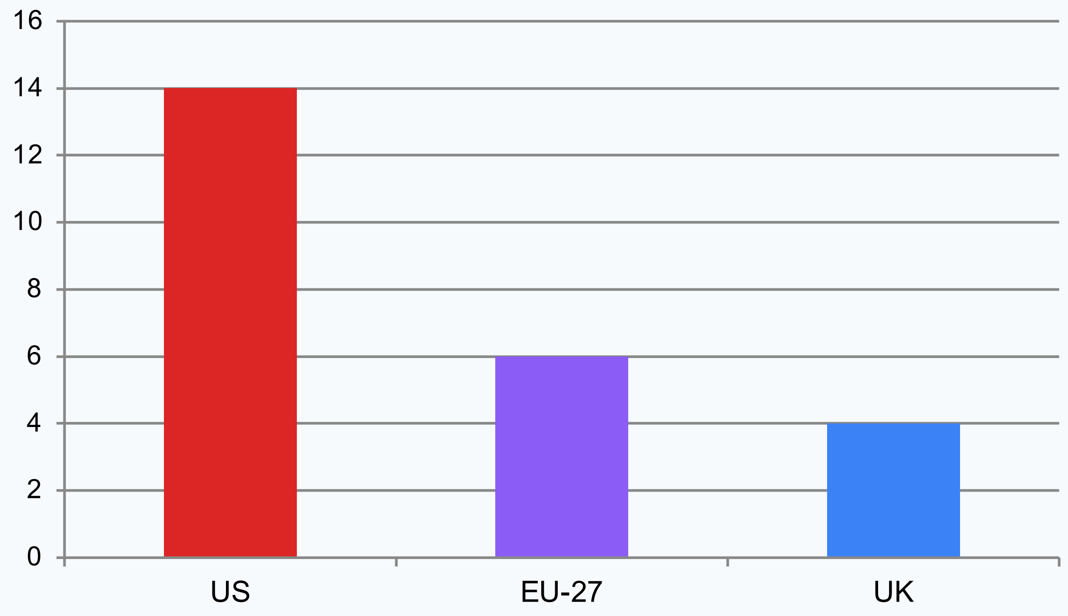
Blue = declines, Orange = increases

Pattern in 2024

Several mature economies reduced emissions, while increases in parts of Asia, the Middle East, and Africa offset part of those gains.

Germany or the US

Among wealthy economies, policy and power-sector composition drive large per-capita differences.



COMPARATIVE BAR / ENERGY-MIX EXPLANATION PANEL

Qualitative message from source:
- France, UK, Portugal: lower per-capita emissions
- Germany, Netherlands, US: higher per-capita emissions
- Key driver: differing shares of nuclear/renewables vs fossil fuels in electricity

Policy implication: prosperity does not mechanically require high emissions; system design choices matter.

Key Takeaways: Responsibility is Shared but Unequal

1. Global annual CO_2 emissions exceed 35 Bt and have not yet peaked.

2. China is the largest annual emitter (~31%), but per-capita emissions remain below the US.

3. Historical responsibility is concentrated in Europe and the US, whose share fell from >90% (1900) to <1/3 today.

4. Per-capita emissions remain deeply unequal, with ~150x gaps between lowest- and highest-emitting countries.

5. Energy policy and technology choices can decouple prosperity from emissions intensity.

Source: Global Carbon Budget 2025 via Our World in Data (fossil fuels and industry CO_2).